



architecture. With the transition to electric powertrains, the architecture of the EV is shrouded in uncertainty. Questions about the number of inputs and outputs, the multitude of sensors, and the organisation of subsystems make designing a unified system daunting. Industry experts shed light on the future of control units in EVs. They explore the necessity of understanding the vehicle's architecture before developing control systems, the importance of data management and privacy, and the feasibility of consolidating control units into a single package.

One of the fundamental challenges in the EV industry is the lack of clarity regarding the architecture of future vehicles. Without a clear understanding of the vehicle's architecture, it is impossible to develop a regular set of standard systems that will work seamlessly. In traditional automotive design, various components and systems operate with proprietary protocols and interfaces, making integration a formidable task.

AUTOSAR, as discussed in a previous article, is instrumental in addressing this issue by providing a standardised framework for automotive software development. However, to leverage AUTOSAR effectively, automakers must first establish a well-defined vehicle architecture.

Decentralised control systems: A pragmatic approach

Given the uncertainty surrounding future vehicle architectures, experts advocate for a flexible vehicle system that can accommodate a wide range of inputs and outputs. This approach involves designing a hardware platform that integrates auto stacks, ensuring compatibility without sacrificing flexibility.

In many cases, a single control system cannot manage the vast number of subsystems that may exist within a vehicle. Consequently,

Revving up quality with ASPICE

ASPICE, or automotive software process improvement and capability determination, is a framework and standard designed to assess and improve the software development processes within the automotive industry.

- **Primary goal.** To ensure high-quality automotive software development and management processes. This includes enhancing product quality, reducing time-to-market, and providing the safety and reliability of automotive software. It is particularly crucial in the context of vehicle software's increasing complexity and critical nature, such as in advanced driver-assistance systems (ADAS), infotainment, and vehicle control systems.
- **Application and benefits.** Companies within the automotive supply chain use ASPICE to evaluate and improve their software development capabilities. By adhering to ASPICE, organisations can achieve better process control, risk management, and product quality. Certification against ASPICE standards is often a requirement for suppliers by major automotive manufacturers (OEMs).
- **Challenges.** Implementing ASPICE can be challenging, especially for smaller organisations, due to the need for significant process changes, training, and possibly cultural shifts within the organisation.

However, the long-term benefits of product quality, customer satisfaction, and market competitiveness often outweigh these initial challenges. ASPICE plays a critical role in shaping the quality and reliability of software development in the automotive industry.

decentralised control systems have emerged as a solution. Multiple control units can oversee specific groups of subsystems, allowing for greater adaptability to evolving architectures.

As EVs become more interconnected and data-driven, data privacy and management concerns are paramount. In Europe, stringent data protection regulations, such as GDPR (general data protection regulation), govern the use of personal data, including data collected from vehicles. These regulations demand explicit consent from vehicle owners for data usage and strict controls on data monetisation.

In contrast, data privacy regulations in India are less stringent. However, experts argue that it is essential for both organisations and governments to establish minimum standards for data privacy and cybersecurity in the EV ecosystem. Cybersecurity norms must be implemented, especially in autonomous driving scenarios where communication is vital.

Collaboration among automakers, technology providers, and governments is crucial to navigating this complex landscape successfully. By establishing clear vehicle architectures, implementing robust data management practices, and carefully

considering control unit consolidation, the EV industry can usher in a new era of sustainable and interconnected transportation. As technology advances, these discussions will undoubtedly shape the future of electric vehicles and mobility.

The role of vehicle control units, adherence to safety standards like ISO 26262, and classification under ASIL categories are essential for seamless integration and global objectives. AUTOSAR, with its vendor neutrality, insurance-like stacks, and ESP certification, revolutionises electric vehicle development, fostering compatibility and competition.

Understanding vehicle architecture and adopting flexible, decentralised control systems are crucial for navigating evolving designs, and data management and privacy considerations are necessary for compliance. As the automotive industry progresses, transparent vehicle architectures, robust data practices, and collaborative efforts will shape the future, paving the way for a sustainable and interconnected transportation landscape. **EFY**

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